AIML PROJECT REPORT

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Project Report: Movie Recommendation System

1. Introduction

In today's digital world, users are exposed to a vast number of movies across different genres. Choosing what to watch can be overwhelming, especially when users are unsure of available options. A movie recommendation system helps simplify this process by suggesting movies based on user preferences.

This project focuses on building a beginner-friendly movie recommendation system using Python. The system recommends movies based on the user's preferred genre.

2. Problem Statement

The main problem addressed in this project is:

"How to help users quickly find movies based on their preferred genre without manually searching through large datasets?"

3. Objective

The objectives of this project are:

- To build a simple movie recommendation system using Python
- To allow users to input a genre and receive relevant movie suggestions
- To handle multiple genres per movie (e.g., "Romance Drama")
- To design a clean and modular project structure

4. Dataset Description

The dataset used in this project is a CSV file (movies.csv) containing two columns:

- title – Name of the movie
- genres – One or more genres associated with the movie

Title	Genres
Titanic	Romance Drama
Avengers	Action Sci-Fi
Frozen	Animation Family

Key Features of Dataset:

- Contains multiple genres per movie
- Covers a wide range of genres such as:
    - Action
    - Comedy
    - Romance
    - Thriller
    - Horror
    - Animation
    - Fantasy
- Designed to avoid missing recommendations for common user inputs

5. Approach

The system follows a simple and structured approach:

Step 1: Load Data

- The dataset is loaded using pandas
- Genres are converted to lowercase for consistency

Step 2: Data Cleaning

- Genres are split into lists
    - Example: "Romance Drama" → "romance", "drama"

Step 3: User Input

- User enters a genre (e.g., "romance")

Step 4: Filtering

- The system checks if the entered genre exists in each movie's genre list

Step 5: Recommendation

- Displays a list of matching movies
- Suggests one random movie for quick selection

6. Project Structure

The project follows a modular structure:

```
movie-recommendation/
main.py
data_loader.py
```

recommender.py
utils.py
movies.csv

File Description:

- main.py
  Handles user interaction and runs the program
- data_loader.py
  Loads and preprocesses the dataset
- recommender.py
  Contains logic for filtering and recommending movies
- utils.py
  Displays available genres

7. Key Decisions

Several important decisions were made during development:

- Used CSV instead of database to keep it simple
- Used pandas for easy data handling
- Converted genres into lists for better matching
- Used modular structure for better readability
- Added random suggestion feature for user convenience

8. Challenges Faced

1. Genre Matching Issue

Initially, genres like:

Romance Drama

were converted into:

romancedrama

This caused mismatches when users entered "romance".

Solution:
Split genres into lists using .split()

2. Limited Dataset

The earlier dataset did not cover enough genres, leading to:

- "No results found" errors

Solution:
Expanded dataset to include:

- More movies
- More genre combinations

3. Handling Multiple Genres

Movies belong to multiple genres, making filtering slightly complex.

Solution:
Used list-based matching:

genre in movie_genres

9. What I Learned:

Through this project, I learned:

- Basics of Python project structuring

- Working with CSV files using pandas
- Importance of data cleaning
- How recommendation systems work at a basic level
- Handling user input and filtering logic
- Writing modular and readable code

10. Limitations

- Recommendations are based only on genre, not user history
- No advanced techniques like machine learning
- No graphical user interface (GUI)
- Limited dataset compared to real-world systems

11. Future Improvements

This project can be improved by:

- Adding movie-based recommendation like Netflix
- Using machine learning algorithms
- Expanding dataset further
- Building a web or mobile interface
- Adding user ratings and reviews

12. Conclusion

This project successfully demonstrates a basic movie recommendation system using Python. It provides users with quick and relevant movie suggestions based on their preferred genre.

Although simple, the project builds a strong foundation for understanding more advanced recommendation systems and real-world applications.

THANK

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